

Baby Bio Written Case Solutions

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When addressing the challenging data that needs to be consolidated from multiple research labs who have different computer systems and/or data in different formats, I can offer my data integration solution. To add to this project, I have decided that the research labs may want to employ hospital personnel to upload data to the database to enhance their analytics. The main idea to aid this case is using a cloud-based data integration platform. A cloud-based platform gives way for multiple hospitals to upload the data in the various formats they use. After their data is uploaded, the platform can standardize, merge and store their data in a centralized database. After successful storage, research labs will have access to the consolidated data analytics.

To integrate the formatting, a data, ingestion API tool can be used. We can create a (Representational State of Transfer) RESTful API for accepting data uploaded by the hospitals in different formats. This API can transform incoming data, handle validating it and enhancing authentication. The second step we need to assist with the various data formats is a data transformation engine. The engine can also standardize the data into a common format. In the process, additional data quality checks, normalization and data cleansing can be performed. Storing consolidated data and reducing data duplication can be done with a centralized database. We can design a relational database using PostgreSQL, NoSQL database or Mongo DB for storage of the consolidated data. Thirdly, we will need to develop a user interface so the hospitals can login and securely upload their data and track the upload status. Once all this is set up, we will need to enable research labs to access the consolidated database by generating reports to gain insights. Two examples of integrative data analytics tools that can work well are Power BI and Tableau.

What we can expect when we can see all the hospitals data standardized into a common format is the ease of ability to compare and analyze the data. This type of centralized data improves data management and reduces data duplication. This organization improves insights greatly, especially with the generated reports. The insights gained can improve decision making and essentially integrate patient care through conclusions presented to the hospitals from the research labs in quarterly or yearly meetings. The scalability of my solution can be designed horizontally so that the platform can handle the large volumes of data from the multiple hospitals and labs. The platform also offers high security to meet growing demand, especially with patient data and their regulatory requirements to comply with the HIPAA laws.

A multi-tenant database architecture can be used for addressing the sorting of proprietary data from hospitals on a specific-case basis. This process will still prioritize that only authorized personnel from those hospitals can access the data. The purpose of this multi-tenant database will provide each hospital it's own access to the stored data. The first step to set this up is using a schema, which will include an identifier for each tenant. A hospital ID can be used for each collection or table and then we need to implement Row-Level Security. RLS will ensure that each hospital will only have access to its own data. Access Control Lists can configure the access or we can use Role-Based Access Control, which will restrict access by only allowing authorized personnel from each hospital to use it. All the consolidated data from each hospital will be stored in a separate table or collection, and when it is accessed through the authorized personnel, they can see the aggregated data.

Since each hospital will have their individual access, creating custom dashboards for each hospital will ensure that the proprietary data will remain confidential. With aggregated data, and metrics for all the hospitals, extra security measures can also be implemented. Using encryption for data at rest and when it is in transit will provide that. The other thing we can do is regularly audit the access logs. This will detect any suspicious activity. From there, we can respond to any potential incidents that may be flagged as threats.

The technical architecture that we will need to design will provide access to the consolidated data through a web service. The web service will have an API gateway such as AWS API Gateway or Azure API Management. Using this web service layer, will limit rates and catching, manage incoming requests and handle authentication. As I mentioned prior, a RESTful API can expose consolidated data to re-search labs. (Create, Read, Update, Delete) CRUD can handle the API operations on the data. NoSQL can ensure the database is optimized for analytics and/or query and store the data once it's consolidated. With oh OAuth or JWT, we can ensure the authorized use of access for the research labs through the web service. Restricted access for specific operations and to data can be done with the authorization mechanism RBAC.

Furthermore, the web service needs to be hosted on a cloud platform such as Google Cloud. In doing so, we can ensure optimal performance, scalability, and reliability. Containerization like Docker can package the web service and provide consistency across environments. To interact with the web service, applications like React and Angular, can be used to provide access to the consolidated data in a user-friendly way with interfaces for the research labs. To interact directly with the web service, research labs can also use an API client like

Postman.

While it is possible, a desktop application would not be recommended. The web-based application provides an ease of maintenance, more flexibility and scalability. The technical architecture with the web based application is also more secure, reliable, and scalable. By creating the system in this way, the research labs can have consolidated data generated by Baby Bio.

To address the billing analytics, a robust metering system shall be created. The usage-based billing will include audit logging to track metrics such as the user ID, date of access, number of times consolidate data is accessed, number of reports generated, the type of activity, the amount of data transferred or processed, number of file uploads and amount of data stored. Aggregation data will also need to be collected by each research lab, with specific time periods from each user. A billing engine like Zuora can calculate the charges based off the billing cycles, pricing tears, and user metrics. The information that we would need to bill our clients incorporates their contact details like names and phone numbers, their billing, address, and a pricing plan choice. Next, we need to collect and store the usage data for each lab The data features time stamps for each usage event and metered metrics usage including: generated reports, which data was accessed and file uploads. Lastly, the billing rules and pricing need to be defined. Billing cycles need to be agreed-upon along with the payment terms and methods, and as I have mentioned prior, pricing plans.

More analytics and reporting also need to be provided. Each research lab should have

access to their usage analytics. That embodies trends, metrics payment, history, and billing. Reporting can also be customizable by offering the capabilities for each lab to choose specific reports for their company. To integrate with existing systems, we can interact with the company's CRM system and manage contacts and information from there. A payment gateway like Google Pay will be able to facilitate this and provide secure payment processing. With all this in place, research, labs will have transparent usage information, the ability to gain insights into usage trends and generate accurate bills.

The internal department will require a data analytics platform to address the analytics and data science needs. Given that the data analytics platform will need to handle various sources of data in large amounts, it will involve providing a data lake. The most popular data lake is JSON and it will be able to process the raw data from the various formats of each lab and store it. Apache NiFi Is an example of a data indigestion tool that can transform, collect and load data into the data lake. For a processing engine, we can use Apache Spark, or Hadoop. These systems can analyze the large data sets involved.

A quality data framework can standardize and clean the data. Additionally, it is able to handle the inconsistencies in data formats to ensure consistency across all data sets. We will need to design a data warehouse, using Amazon Redshift or Google BigQuery. This will store the processed and aggregated data. Business intelligence tools like Power BI, Will bring us reports and interactive visualizations. Integrating machine learning like TensorFlow will help us identify patterns in the data and build predictive models. Data governance framework can establish compliance with regulatory requirements, and ensure security and data quality. Applying RBAC

will restrict access to sensitive data and analytics.

Harnessing the cloud infrastructure, will leverage our ability to store and process data according to volume and scale using compute resources. To process the large data sets in parallel, we can utilize distributed computing frameworks like Apache Spark. Employing this architecture, our internal department can analyze the large data volumes, process them and provide valuable insights to the research labs. We can also drive businesses by increasing proficient decision-making through predictive models.